



אוניברסיטת בן-גוריון בנגב

Ben-Gurion University of the Negev

הפקולטה למדעי ההנדסה

המחלקה להנדסת חשמל ומחשבים

Faculty of Engineering Science

Dept. of Electrical and Computer Engineering

פרויקט הנדסי שנה ד'

Fourth Year Engineering Project

Final Report

Neural Brain Fields for EEG Reconstruction with 3D Gaussian Splatting

Project number: P-2026-144	מספר הפרויקט: P-2026-144
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Submission date: 21.12.25	תאריך הגשה: 21.12.25

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תקציר

פרויקט זה עוסק בשחזור ערוצי EEG חסרים באמצעות ייצוגים מבוססי Gaussian Splatting. נקודת המוצא המחקרית הייתה שיטת Neural Brain Fields, אך במימוש הסופי פותחה מערכת ייעודית המבוססת ישירות על ייצוגי גאוסיאנים במרחב ובמרחב-זמן. לצורך הערכה אובייקטיבית הוסתרו בכל נבדק חמש אלקטרודות אמיתיות, המודל אומן על 55 האלקטרודות הנותרות, והאותות המשוחזרים הושוו לאותות המקוריים הידועים. תחילה בוצעה השוואת פונקציות הפסד על נבדק S01, שבעקבותיה נבחר MSE כיעד האימון הקבוע. לאחר מכן בוצעה השוואה מבוקרת בין 3D GS ל- 4D GS על 18 נבדקים. במוצע, 4D GS השיג PCC גבוה יותר (0.3567 לעומת 0.3455), שגיאות MAE ו-RMSE נמוכות מעט יותר, ושיפור בולט יותר בשימור המשרעת, כאשר יחס ה-STD עלה מ-0.4002 ל-0.4229. עם זאת, השיפור אינו אחיד בין נבדקים ואלקטרודות, ואלקטרודה 57 התנהגה כמקרה חריג. התוצאות מצביעות על יתרון ממוצע מתון לייצוג 4D ועל פוטנציאל להמשך שיפור, אך אינן מצדיקות טענה לעליונות עקבית בכל תנאי.

מילות מפתח: EEG, שחזור ערוצים חסרים, אלקטרודות וירטואליות, 4DGS, 3D-GS, Gaussian Splatting.

Abstract

This project investigates missing-channel reconstruction in electroencephalography (EEG) using Gaussian Splatting based representations. Neural Brain Fields (NBF) provided the initial research inspiration, but the final implementation was developed as a direct EEG-specific Gaussian representation rather than an NBF-GS hybrid. A leave-electrodes-out protocol was used for quantitative validation: five recorded channels were excluded from training, the models were optimized using the remaining 55 channels, and the reconstructed signals were compared with the known recorded ground truth. A preliminary loss-function study on subject S01 selected standard mean squared error (MSE) as the fixed training objective. The final controlled experiment then compared 3D Gaussian Splatting (3D-GS) and a spatiotemporal 4DGS variant on 18 subjects using identical hidden electrodes, preprocessing, input recordings, and evaluation metrics. Across subjects, 4DGS achieved a modest mean advantage: PCC increased from 0.3455 to 0.3567, MAE decreased from 8.10 to 8.03 μV , RMSE decreased from 10.90 to 10.82 μV , and STD ratio improved from 0.4002 to 0.4229. The strongest improvement was therefore in amplitude preservation. The gain was not uniform across subjects or electrodes, and electrode 57 remained a challenging case. The final system demonstrates a reproducible Gaussian-based EEG reconstruction pipeline and supports 4D modeling as a promising, but not universally superior, extension of the 3D baseline.

Keywords: EEG, missing-channel reconstruction, virtual electrodes, Gaussian Splatting, 3D-GS, 4DGS, spatiotemporal modeling.

1 Introduction

1.1 Motivation

Electroencephalography (EEG) measures electrical potential at discrete scalp locations with millisecond-scale temporal resolution. Its main spatial limitation is that only a finite set of electrodes samples a smooth, volume-conducted electrical field. As a result, information between electrodes is not measured directly, and damaged, disconnected, or intentionally omitted channels can reduce the spatial information available for subsequent processing. The physical smoothness and spatial correlation of scalp potentials motivate reconstruction methods that estimate signals at locations where no measurement is provided [1].

Traditional interpolation can fill missing channels using geometric assumptions, while neural methods can learn signal structure from data. Recent work on virtual EEG electrodes demonstrated that convolutional networks can reconstruct or upsample channels and can outperform classical interpolation in some settings [2]. EEG-specific deep networks such as EEGNet and deeper convolutional architectures further show that learnable spatial-temporal filters can capture meaningful structure in EEG signals [3], [4]. These methods, however, usually operate on a fixed channel layout and do not explicitly represent the measured activity as a continuous field.

1.2 From Neural Fields to Gaussian Representations

Neural Radiance Fields (NeRF) introduced the idea of representing a continuously queryable scene through a learned function rather than a fixed discrete grid [5]. Neural Brain Fields (NBF) transferred this concept to EEG by learning a continuous mapping from spatial coordinates and time to voltage, enabling reconstruction at unmeasured positions [6]. NBF was the main conceptual starting point of this project: instead of viewing an EEG recording only as a matrix of channels by time, it can be viewed as sparse samples from an underlying spatiotemporal field.

The final engineering direction changed from the preliminary proposal. Rather than implementing an NBF network followed by a separate Gaussian renderer, the project developed EEG-specific Gaussian representations directly. This choice emphasized local spatial basis functions, differentiable learning, and controlled hidden-channel reconstruction. The final comparison focuses on a 3D Gaussian baseline and a 4D spatiotemporal extension while holding the loss function and validation protocol fixed.

1.3 Final Project Objective

The final objective is to determine whether Gaussian representations can reconstruct intentionally hidden EEG channels and whether adding a stronger spatiotemporal representation in 4D-GS provides a measurable advantage over the 3D-GS baseline. The project therefore treats missing-

channel reconstruction as the primary product function and quantitative validation against real held-out electrodes as the primary acceptance criterion.

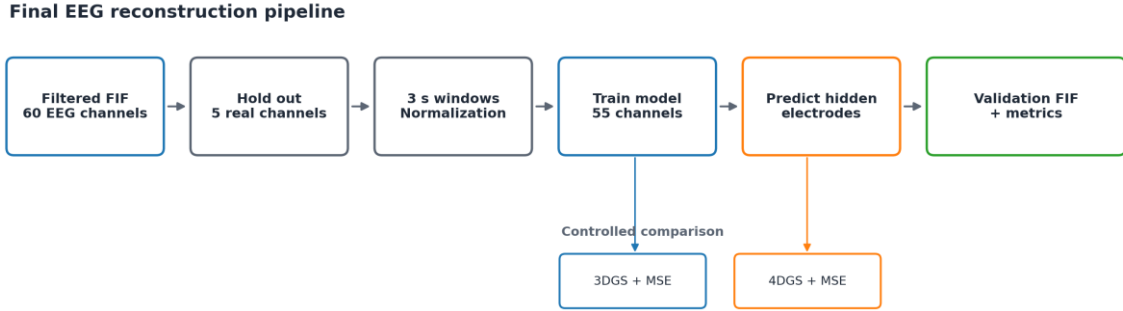


Fig. 1. Final EEG reconstruction and validation pipeline. The same filtered recordings, five held-out electrodes, MSE objective, and evaluation metrics are used for both representations.

1.4 Main Contributions

- An end-to-end EEG reconstruction pipeline that loads filtered FIF recordings, preserves original channels, trains a Gaussian field on retained electrodes, and writes reconstructed validation channels to a separate FIF file.
- A leave-electrodes-out validation protocol using five real electrodes per subject, eliminating the need to assess synthetic electrodes without ground truth.
- An EEG-adapted 3D Gaussian field with learnable spatial scales and time-dependent amplitudes, followed by a controlled 4D-GS extension.
- A loss-function ablation on S01 that established standard MSE as the most reliable objective among the tested variants.
- A paired 3D-GS versus 4D-GS evaluation on 18 subjects, including overall, subject-level, electrode-level, amplitude-preservation, and outlier analyses.

1.5 Report Organization

Section 2 summarizes the technical background and related work. Section 3 states the final system specification and changes from the preliminary report. Section 4 presents the chosen engineering approach and mathematical model. Section 5 defines the final acceptance tests and measurement methodology. Section 6 reports and interprets the experimental results. Section 7 describes major engineering problems and solutions. Section 8 concludes the project, compares planned and achieved objectives, and gives recommendations for future work. Reproducibility information and the required blank evaluation form are provided in the appendices.

2 Background and Related Work

2.1 EEG Spatial Reconstruction

EEG electrodes measure scalp potentials generated by distributed neural sources after propagation through brain tissue, cerebrospinal fluid, skull, and scalp. Volume conduction smooths these fields spatially, producing substantial correlation between nearby electrodes while also limiting spatial resolution [1]. A missing channel therefore does not imply that no information exists about its signal; neighboring channels and temporal structure can provide predictive information. The engineering challenge is to exploit that redundancy without introducing unrealistic temporal waveforms or excessively attenuating signal amplitude.

Classical spherical-spline interpolation uses electrode geometry and smoothness assumptions. Learned reconstruction methods can additionally exploit temporal patterns and nonlinear relationships. Svantesson et al. showed that convolutional encoder-decoder networks can generate virtual EEG electrodes and evaluated them using amplitude, correlation, coherence, phase, and distance measures [2]. This establishes missing-channel reconstruction as a measurable supervised task when true withheld channels are available.

2.2 Deep Learning for EEG

EEGNet uses compact temporal and depthwise spatial convolutions to learn physiologically interpretable features with relatively few parameters [3]. Deeper convolutional networks can also learn spectral and spatial features directly from EEG and have been analyzed using feature visualization techniques [4]. These works motivate learned spatial-temporal processing, but their output remains tied to a predefined electrode arrangement. In contrast, the present project attaches model inputs to spatial coordinates, allowing the reconstruction problem to be expressed as estimating voltage at a location and time rather than predicting a fixed channel index.

2.3 Neural Radiance Fields and Neural Brain Fields

NeRF represents a three-dimensional scene as a continuous neural function that is queried at arbitrary coordinates [5]. Its key conceptual advantage for this project is not image rendering itself, but continuous representation: the learned function is not restricted to a fixed set of sampled points. NBF applies this idea to brain recordings and models EEG as a continuous four-dimensional field of position and time [6]. The NBF manuscript demonstrates that unseen electrode positions can be reconstructed from a single recording and uses non-overlapping 3 s windows in its processing pipeline.

NBF therefore provided two important ideas used in this project: first, represent each EEG measurement together with its physical coordinates and time; second, validate the model by querying

it at electrode locations withheld from training. The final implementation departs from NBF by replacing an implicit positional-encoding MLP with Gaussian basis functions tailored to the electrode geometry.

2.4 3D Gaussian Splatting

3D Gaussian Splatting (3D-GS) represents a scene using localized anisotropic Gaussian primitives rather than repeatedly querying a large implicit network [7]. In the original computer-vision formulation, each Gaussian has a position, covariance, opacity, and appearance parameters, and the representation is optimized differentially. The explicit local support of Gaussians is attractive for EEG because electrodes are sparse spatial samples and nearby locations should influence one another more strongly than distant locations.

The project does not reproduce the graphics rasterization pipeline of the original 3D-GS paper. Instead, Gaussian Splatting is adapted as a learnable continuous interpolation field over electrode coordinates. The term 3D-GS in this report therefore refers to the project-specific spatial Gaussian EEG representation, while [7] is the methodological origin of the Gaussian representation concept.

2.5 4D Gaussian Modeling for EEG

Dynamic signals motivate extending Gaussian representations so that time participates directly in the learned representation. During the project, the emerging EMAG work proposed differentiable 4D Gaussian mixture splatting for EEG spatial super-resolution [8]. EMAG represents electrical sources as anisotropic space-time Gaussians and renders them to scalp electrodes. It provides independent evidence that Gaussian models can be generalized from static 3D scene representations to dynamic EEG fields.



Fig. 2. Related-work schematic of EMAG 4D Gaussian EEG super-resolution, adapted from [8]. This figure is included only as research context; it is not the architecture or a result of the present project.

The project 4D-GS implementation was developed independently for the hidden-channel reconstruction pipeline and should not be interpreted as a reproduction of EMAG. In particular, this report does not attribute EMAG's full 4D precision-matrix parameterization, spherical source grid, or published performance to the project. The EMAG paper is used only as a related-work reference supporting the broader motivation for explicit spatiotemporal Gaussian modeling.

2.6 Research Gap

The practical gap addressed here is narrower than general EEG super-resolution: can an explicit Gaussian field reconstruct real channels that are deliberately withheld, and does the addition of a 4D representation improve reconstruction when all other experimental conditions remain the same? Existing literature establishes continuous neural fields, learned virtual electrodes, and Gaussian representations separately. The present project contributes a controlled engineering evaluation of 3D-GS and 4D-GS under the same EEG missing-channel protocol.

3 Final Technical Specification

3.1 Final Product Definition

The final product is a software pipeline for EEG missing-channel reconstruction. It accepts a preprocessed EEG recording in FIF format, intentionally excludes a specified subset of real electrodes from the training set, optimizes a Gaussian EEG model on the remaining channels, predicts voltage at the hidden electrode positions, and stores the predictions as new validation channels in a separate FIF file. The ground-truth channels remain present and unchanged so that reconstruction quality can be measured directly.

Table 1. Final technical specification and verification method.

Specification item	Final value / behavior	Verification
Input format	Filtered MNE FIF recording with 60 EEG channels	File loading and channel-count checks
Sampling rate	120 Hz in the processed Brennan recordings	Read from FIF metadata
Windowing	Sequential 3 s windows (360 samples per channel)	Window index and time-range logging
Validation channels	17, 11, 52, 57, 19	Excluded from all training batches
Training channels	55 original electrodes	Channel-selection check before model creation
Model variants	3D-GS baseline and 4D-GS extension	Separate paired output files
Final loss	Mean squared error (MSE)	Fixed for both models after S01 ablation
Output	Original 60 channels + VAL_17, VAL_11, VAL_52, VAL_57, VAL_19	Separate validation FIF
Core metrics	PCC, R^2 , MAE [μ V], RMSE [μ V], STD ratio, RMS ratio	Direct prediction-to-ground-truth comparison
Data integrity	Original channels must remain numerically unchanged	Reload-source and max-difference checks

3.2 Changes from the Preliminary Report

The preliminary proposal intentionally defined a broader research program. As implementation and validation progressed, the scope was narrowed to the most measurable and technically defensible part of the project. The final report therefore documents both the original plan and the actual implemented system rather than presenting removed goals as completed work.

Table 2. Changes from the preliminary plan to the final project scope.

Preliminary plan	Final implementation	Reason / implication
Hybrid NBF + 3D-GS architecture	Direct EEG-specific Gaussian field; NBF retained as conceptual inspiration	Reduced architectural complexity and focused the project on Gaussian reconstruction itself
Primary baseline: NBF	Primary final comparison: 3D-GS versus 4D-GS	Allowed a controlled representation comparison under identical loss and hidden channels
Real-time 3D visualization	Removed from final acceptance scope	Validation effort prioritized quantitative hidden-channel reconstruction
EEG-to-image / visual-stimulus reconstruction	Removed from final scope	Too broad for reliable completion within the project schedule
10% reconstruction improvement and 30% latency reduction versus NBF	Replaced by measured 3D-GS/4D-GS metrics; original numeric target reported as not verified	Avoids claiming unmeasured NBF performance or speedup
Virtual electrodes / missing-channel reconstruction	Implemented and evaluated on real held-out channels	Became the central product function

3.3 Final Acceptance Criteria

The final acceptance criteria are based on functionality, data integrity, and measurable reconstruction performance. The system must:

1. Train without access to the five validation channels.
2. Reconstruct the five hidden channels with the correct number of samples.
3. Preserve all original FIF channels exactly, without modifying their values.
4. Compute quantitative evaluation metrics using the denormalized reconstructed signals.

Model quality is assessed comparatively rather than by applying an arbitrary single correlation threshold. The original preliminary targets that were removed from the revised final scope are evaluated explicitly in Section 8 rather than being omitted.

4 Chosen Approach and Engineering Design

4.1 Data Flow and Software Architecture

The system is organized as a sequence of reproducible stages. First, the filtered recording and electrode geometry are loaded with MNE-Python [10]. Second, the five validation electrodes are removed only from the training view of the data. Third, the remaining measurements are converted to coordinate-time samples and normalized. Fourth, the selected Gaussian model is optimized window by window. Fifth, the model is queried at the spatial positions of the hidden electrodes for every sample. Finally, predicted channels are denormalized, appended to a fresh copy of the original recording, and evaluated.

Leave-electrodes-out validation protocol

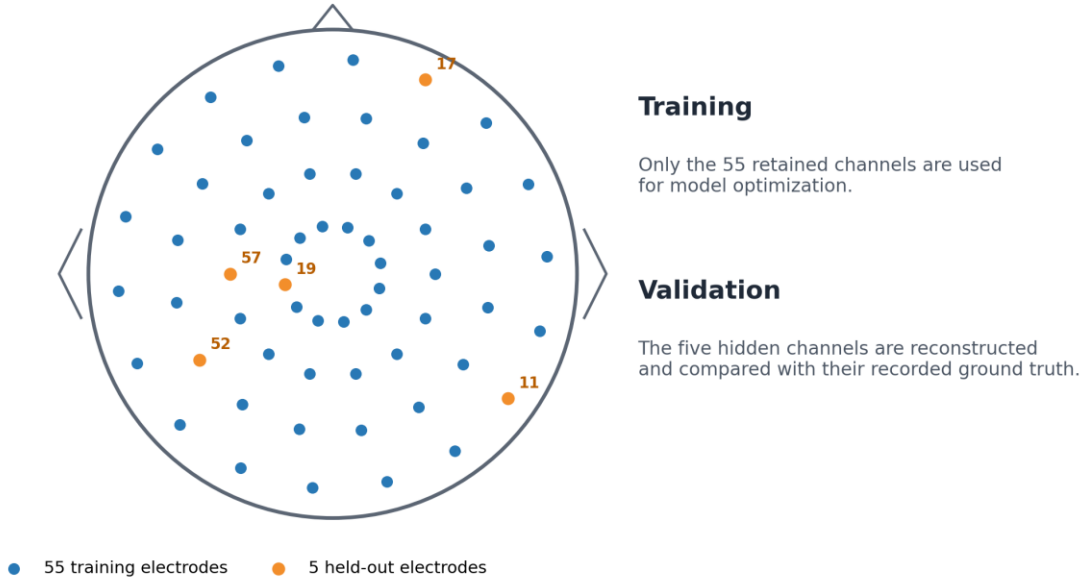


Fig. 3. Leave-electrodes-out validation protocol used in the project. The electrode placement is schematic; the five labels correspond to the actual hidden channels.

4.2 Dataset and Preprocessing

The experiments use the Brennan naturalistic-listening EEG dataset introduced by Brennan and Hale [9] and processed in the NBF work [6]. The processed dataset contains 60 EEG sensors and was downsampled to 120 Hz with a 0.5-59.5 Hz band-pass preprocessing pipeline in the NBF data preparation [6]. The underlying experiment contains passive listening to a 12.4-minute audiobook [9]. The final 3D-GS-versus-4D-GS analysis includes the 18 subjects for which paired outputs from both model variants were available.

The processing pipeline uses 3 s windows. At 120 Hz, each complete window contains 360 time samples per retained electrode. Each electrode-time sample is represented by a four-element input vector $[x, y, z, t]$ together with the measured voltage. Spatial coordinates are taken from the electrode montage associated with the recording, and absolute window time is maintained so that predictions are written back in the correct temporal order.

4.3 Coordinate and Voltage Normalization

Normalization is performed to keep spatial, temporal, and voltage values in numerically stable ranges. In the 3D-GS implementation, x , y , and z coordinates are min-max normalized to $[-1, 1]$, time is normalized to $[0, 1]$ inside the working representation, and voltage is z-score normalized for optimization. The prediction is transformed back to physical voltage units before evaluation and FIF export. The same preprocessing and input data are used in the final 3D-GS and 4D-GS comparison, so the measured differences primarily reflect the model representation rather than a changed normalization pipeline.

4.4 3D-GS EEG Field

The verified 3D-GS baseline uses one Gaussian center for each unique training-electrode position. Let μ_i denote the three-dimensional center of Gaussian i and s_i its positive axis-wise spatial scale. The implementation stores unconstrained scale parameters and converts them with a softplus transform, which guarantees positive widths. A positive amplitude factor α_i is applied to each spatial Gaussian. For a query position $\mathbf{x} = [x, y, z]$, the unnormalized spatial contribution is

$$g_i(\mathbf{x}) = \alpha_i \exp \left[-\frac{1}{2} \sum_d \left(\frac{x_d - \mu_{i,d}}{s_{i,d}} \right)^2 \right] \quad (1)$$

The spatial contributions are normalized across the K Gaussian centers so that the model forms a location-dependent mixture:

$$w_i(\mathbf{x}) = \frac{g_i(\mathbf{x})}{\sum_j g_j(\mathbf{x})} \quad (2)$$

Time is modeled by a compact multilayer perceptron with SiLU activations. It maps the scalar time value to K time-dependent amplitudes $a_i(t)$, one for each Gaussian center. The reconstructed voltage is the weighted sum

$$\hat{V}(\mathbf{x}, t) = \sum_i w_i(\mathbf{x}) a_i(t) \quad (3)$$

This architecture is substantially smaller and more localized than a generic Fourier-feature neural field. The final implementation does not use Fourier positional encoding; coordinates are passed directly to the Gaussian field. Gaussian centers are initialized from the training electrode positions, while their learnability can be controlled by a model option. The temporal network hidden dimension is 64 in the project configuration, and optimization uses Adam.

4.5 4D-GS Extension

The 4D-GS variant was introduced after the 3D baseline and loss-function experiments. Its purpose is to give time a more direct role in the learned Gaussian representation, rather than relying only on a spatial Gaussian mixture with temporal amplitude modulation. The final comparison treats 4D-GS as a spatiotemporal extension of the same EEG reconstruction task. Importantly, the report does not equate the project implementation with the specific full 4D precision-matrix architecture of EMAG [8], because that formulation was used as research reference rather than copied as the project model.

The prediction code never treats generated channels as substitutes for the original data. After inference, the original FIF file is reloaded from disk as the source of truth, the predicted channels are created separately, and only those new channels are appended. The output therefore contains the 60 original channels plus five validation channels. Integrity checks compare the original channel arrays before and after output construction and reject sample-count mismatches or duplicate channel names.

This safeguard was added because an early development risk was unintended modification or overwrite of measured signals during channel generation.

Project Gaussian representations

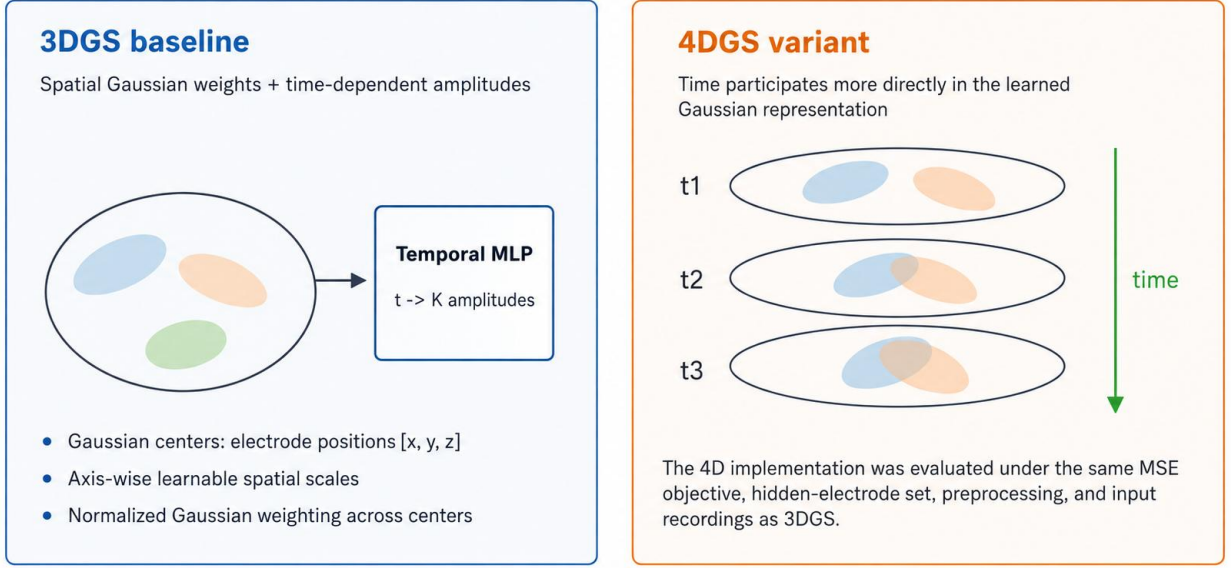


Fig. 4. Project Gaussian representations. The left panel reflects the verified 3D-GS implementation (spatial Gaussian weighting plus time-dependent amplitudes). The right panel is a conservative conceptual view of the independently implemented 4D-GS variant used for the paired experiment.

4.6 Loss Function Selection

A loss-function study was performed on S01 before the full paired comparison. Four 3D-GS objectives were tested: standard MSE, MSE plus an STD-matching term, Huber plus STD loss, and Huber plus scale regularization and Laplacian smoothing. The purpose was to determine whether amplitude preservation and robustness terms could reduce the visibly attenuated predictions without degrading waveform accuracy. Standard MSE achieved the best PCC, R^2 , MAE, and RMSE among the tested 3D loss variants, while the added STD term improved amplitude ratio but slightly reduced correlation and increased error. MSE was therefore fixed for both 3D-GS and 4D-GS in the final experiment.

$$\mathcal{L}_{\text{MSE}} = \frac{1}{N} \sum_{n=1}^N (\hat{V}_n - V_n)^2 \quad (4)$$

4.7 Validation FIF Generation and Data Integrity

The prediction code never treats generated channels as substitutes for the original data. After inference, the original FIF file is reloaded from disk as the source of truth, the predicted channels are created separately, and only those new channels are appended. The output therefore contains the 60 original channels plus five validation channels. Integrity checks compare the original channel arrays before and after output construction and reject sample-count mismatches or duplicate channel names.

This safeguard was added because an early development risk was unintended modification or overwrite of measured signals during channel generation.

4.8 Engineering Design Considerations

The main design trade-off is between representation flexibility and robustness under sparse sampling. A Gaussian field provides a smooth local basis and is naturally tied to electrode geometry, but it can attenuate amplitudes when the learned mixture averages neighboring signals. Increasing temporal flexibility can improve dynamic behavior, but it also increases model complexity, computation time, and sensitivity to specific electrodes. For this reason, the final experimental design changes one major factor at a time: the MSE objective, hidden electrodes, preprocessing, and evaluation procedure remain fixed while the representation changes from 3D-GS to 4D-GS.

5 Final Acceptance Tests and Experimental Setup

5.1 Test Setup

The acceptance-test setup is entirely data-driven. Each experiment begins from a filtered FIF recording containing all 60 measured channels. Channels 17, 11, 52, 57, and 19 are marked as validation targets and are excluded from model training. Their recorded signals remain available only for post-inference evaluation. The model is trained on the other 55 channels and then queried at the five held-out positions over the complete recording. The resulting five predictions are denormalized and compared sample-by-sample with the corresponding ground truth.

The final 3D-GS-versus-4D-GS experiment uses paired conditions: for a given subject, both models receive the same source recording, the same 55 training channels, the same five validation locations, the same MSE objective, and the same metric implementation. This paired design minimizes confounding from preprocessing or target selection. Eighteen subjects have complete paired outputs and are included in the final analysis.

5.2 Evaluation Metrics

Table 3. Evaluation metrics used in the acceptance tests.

Metric	Meaning	Desired direction
PCC	Pearson correlation between predicted and true temporal waveforms	Higher is better
R²	Explained variance of the true signal by the prediction	Higher is better
MAE [μV]	Mean absolute voltage error after denormalization	Lower is better
RMSE [μV]	Root mean squared voltage error after denormalization	Lower is better
STD ratio	$\text{std}(\text{prediction}) / \text{std}(\text{ground truth})$	Closer to 1 is better
RMS ratio	$\text{rms}(\text{prediction}) / \text{rms}(\text{ground truth})$	Closer to 1 is better

Correlation and R² characterize waveform similarity and explained variance, while MAE and RMSE quantify error magnitude in physical voltage units. STD and RMS ratios were added because early

experiments showed that a model can follow waveform shape while producing amplitudes that are too small. These amplitude ratios therefore complement the error metrics and reveal over-smoothing or attenuation.

5.3 Loss-Function Acceptance Test on S01

The S01 ablation is a design-selection test rather than the final performance claim. All variants use the same five hidden channels and are compared over the full recording. The selected objective should improve overall waveform reconstruction rather than optimizing only one metric. The 4D-GS+MSE S01 result is included as an early reference measurement, but it is not treated as a fifth loss function.

Table 4. S01 loss-function selection results.

Experiment	PCC	R ²	MAE [μ V]	RMSE [μ V]	STD ratio	RMS ratio
MSE	0.4377	0.1909	8.372	11.303	0.481	0.481
MSE + STD	0.4314	0.1692	8.503	11.433	0.564	0.564
Huber + STD	0.3867	0.1405	8.563	11.683	0.469	0.470
Huber + scale + Laplacian	0.3946	0.1542	8.479	11.604	0.429	0.429
4D-GS + MSE reference	0.6081	0.1240	7.406	10.108	0.898	0.899

5.4 Measurement Accuracy and Experimental Limitations

Because the acceptance tests operate on existing digital recordings, there is no additional analog measurement instrument whose calibration uncertainty must be propagated. The dominant sources of evaluation uncertainty are biological variability, subject-to-subject differences, electrode location, preprocessing, normalization, and stochastic model optimization. All final metrics are calculated after restoring voltage units, and the same code path is used for both model variants.

The error bars in the overall result figures represent inter-subject standard deviation, not confidence intervals or repeated-seed uncertainty. The final experiment contains one paired output per model for each included subject and does not establish statistical significance across multiple random initializations. Conclusions are therefore stated as descriptive mean advantages and observed trends rather than as proof of universal superiority. The electrode-57 exclusion analysis is explicitly diagnostic and does not replace the complete five-electrode result.

6 Results and Discussion

6.1 Loss-Function Selection

The S01 results show a clear trade-off between amplitude preservation and conventional reconstruction accuracy. Standard MSE produced the best PCC, R², MAE, and RMSE among the four 3D-GS loss variants. Adding STD matching raised the STD ratio from 0.481 to 0.564, confirming that the additional term increased predicted amplitude, but PCC fell from 0.438 to 0.431 and both

error metrics increased. The Huber-based objectives did not recover this loss in accuracy, and the scale-plus-Laplacian variant produced the weakest amplitude ratio. Standard MSE was therefore selected because it gave the most reliable overall reconstruction rather than the best value on a single amplitude metric.

The early 4D-GS+MSE S01 reference was substantially stronger in PCC and error metrics: PCC increased to 0.608, MAE fell to 7.41 μV , RMSE fell to 10.11 μV , and STD ratio rose to 0.898. However, R^2 decreased to 0.124 because one validation channel, electrode 57, behaved unstably. This single-subject result motivated the full multi-subject comparison rather than being used as the final conclusion.

6.2 Overall 3D-GS versus 4D-GS Performance

Table 5. Overall 3D-GS versus 4D-GS results across 18 paired subjects.

Metric	3D-GS + MSE	4D-GS + MSE	Relative change	Mean winner
Mean PCC	0.3455 ± 0.1270	0.3567 ± 0.1377	+3.24%	4D-GS
Mean R^2	0.1271 ± 0.1301	0.1273 ± 0.1201	+0.20%	4D-GS
Mean MAE [μV]	8.10 ± 2.14	8.03 ± 2.14	0.82% lower	4D-GS
Mean RMSE [μV]	10.90 ± 3.20	10.82 ± 3.21	0.78% lower	4D-GS
Mean STD ratio	0.4002 ± 0.1277	0.4229 ± 0.1699	+5.67%	4D-GS
Mean RMS ratio	0.4003 ± 0.1277	0.4230 ± 0.1700	+5.68%	4D-GS

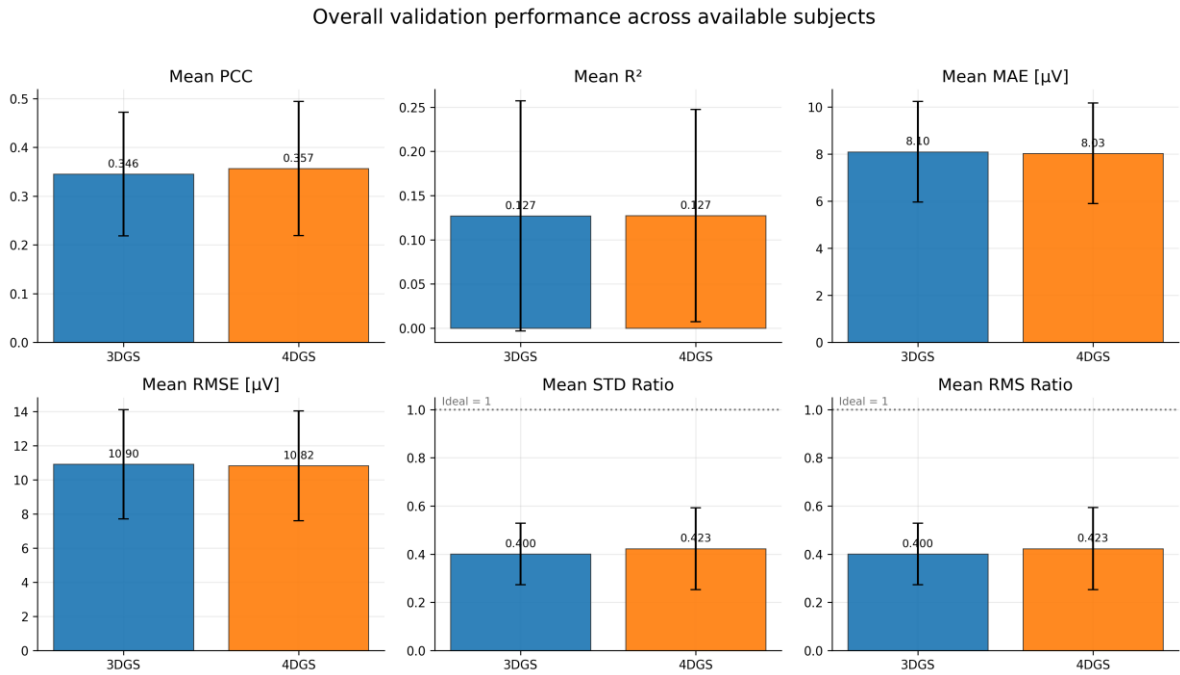


Fig. 5. Overall validation performance across the 18 paired subjects. Error bars indicate inter-subject standard deviation. Blue: 3D-GS; orange: 4D-GS.

All six reported mean metrics favor 4D-GS, but the magnitude of the gain differs substantially by metric. PCC improves by 3.24%, while R^2 is essentially unchanged at the displayed precision. MAE and RMSE improve by less than 1%. The largest relative changes are the STD and RMS ratios, both improving by approximately 5.7%. Therefore, the primary observable advantage of 4D-GS is better preservation of signal magnitude rather than a large reduction in pointwise reconstruction error.

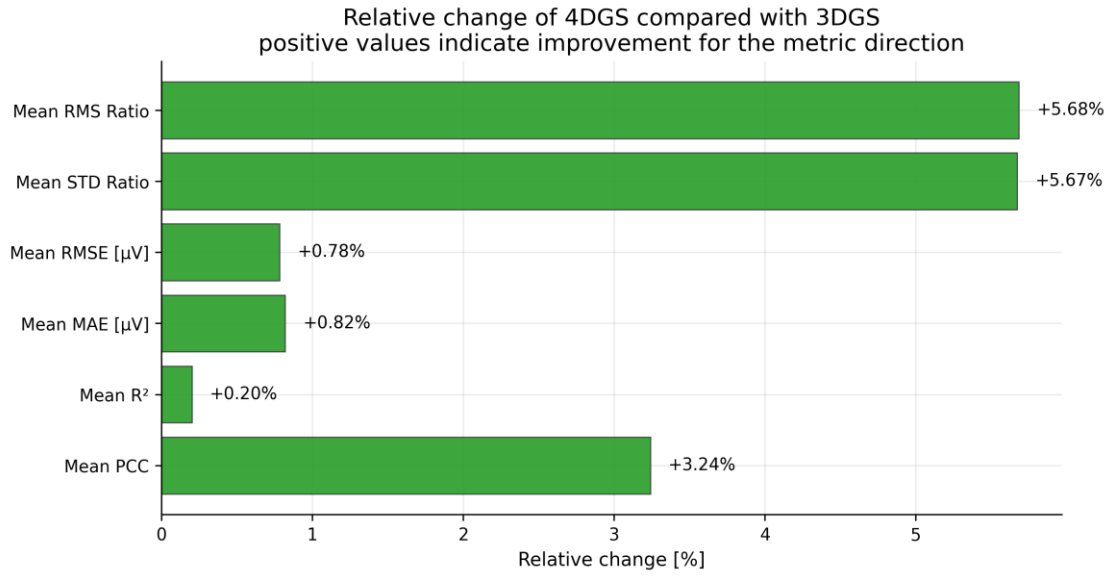


Fig. 6. Relative change of 4D-GS compared with 3D-GS. Positive values denote improvement according to the preferred direction of each metric.

6.3 Subject-Level Behavior

The mean values alone can overstate consistency. At subject level, most 3D-GS and 4D-GS results are close, while S01 shows a substantially larger 4D benefit than the majority of the cohort. For PCC and R^2 , 3D-GS performs better in slightly more individual subjects even though the mean favors 4D-GS. The error and amplitude metrics are more evenly split. Consequently, the final conclusion is a modest average 4D advantage, not subject-by-subject dominance.

Subject-level comparison for key metrics



Fig. 7. Subject-level comparison for PCC, RMSE, and STD ratio. Connectors show the direction and magnitude of the difference between 3D-GS and 4D-GS for each subject.

6.4 Electrode-Level Behavior

Table 6. Electrode-level comparison across the five held-out channels.

Electrode	PCC 3D→4D	R ² 3D→4D	RMSE [μV] 3D→4D	STD ratio 3D→4D
11	0.389 → 0.407	0.161 → 0.183	12.67 → 12.39	0.422 → 0.442
17	0.361 → 0.375	0.139 → 0.153	9.00 → 8.94	0.422 → 0.433
19	0.381 → 0.398	0.162 → 0.182	13.20 → 12.99	0.396 → 0.414
52	0.252 → 0.264	0.047 → 0.071	11.95 → 11.82	0.340 → 0.347
57	0.344 → 0.339	0.127 → 0.047	7.70 → 7.97	0.421 → 0.479

Electrodes 11, 17, 19, and 52 all show the same qualitative pattern: 4D-GS improves PCC and R², lowers RMSE, and slightly increases the amplitude ratio. Electrode 57 is different. Its STD ratio improves from 0.421 to 0.479, but PCC decreases, R² falls substantially, and RMSE increases. In other words, increasing signal strength at this location does not improve the temporal match to the true waveform.

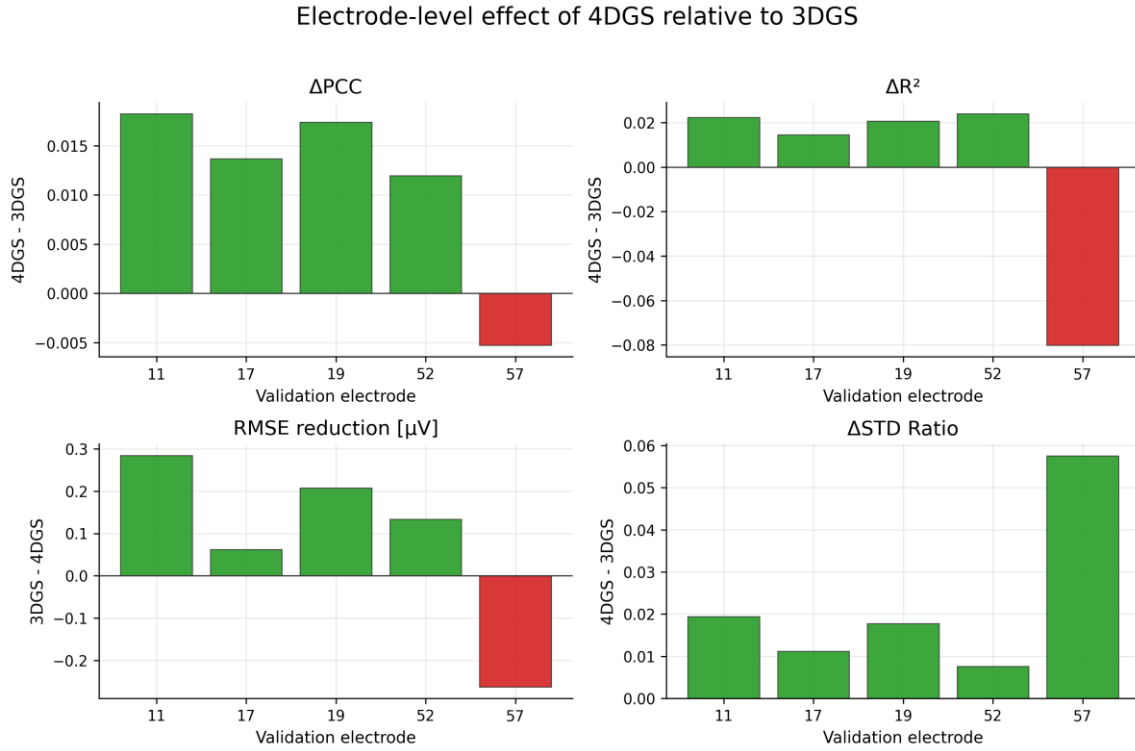


Fig. 8. Electrode-level changes when moving from 3D-GS to 4D-GS. Positive bars represent improvement in the preferred metric direction; electrode 57 is the main exception.

6.5 Sensitivity Analysis Excluding Electrode 57

A secondary analysis was performed without electrode 57 to quantify how strongly this difficult channel affects the aggregate conclusion. This is not the primary result and does not replace the five-electrode validation. Without electrode 57, mean PCC increases from 0.3458 for 3D-GS to 0.3611 for 4D-GS, a relative improvement of 4.43%. Mean R^2 increases from 0.1270 to 0.1474 (16.02%). MAE decreases from 8.6881 to 8.5714 μV and RMSE from 11.7053 to 11.5333 μV . STD and RMS ratios increase from 0.3950 to 0.4090.

This sensitivity check supports two conclusions. First, electrode 57 is genuinely influential and should be investigated rather than ignored. Second, even after its removal, the 4D advantage remains moderate for correlation and error, so the project should not present the result as a dramatic universal improvement.

6.6 Direct Similarity Between 3D-GS and 4D-GS Predictions

The two model variants usually predict very similar waveforms. Across paired validation predictions, the mean PCC between 3D-GS and 4D-GS output signals is 0.9384. The mean absolute difference is 0.8460 μV and the RMSE between predictions is 1.4934 μV . The mean predicted STD rises from 5.0785 μV for 3D-GS to 5.3252 μV for 4D-GS, and mean RMS rises from 5.0788 to 5.3259 μV . This explains why amplitude ratios improve more than waveform metrics: 4D-GS generally modifies the

magnitude of an already similar waveform rather than learning a completely different temporal trajectory.

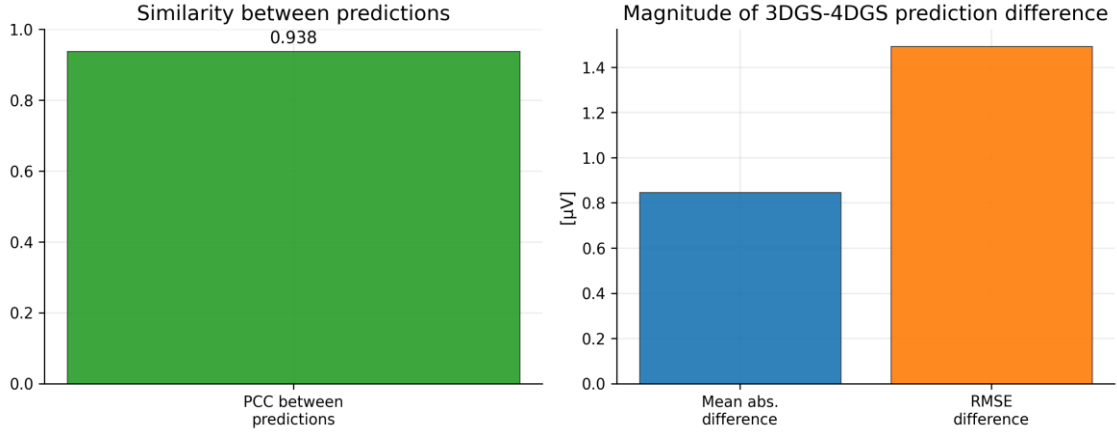


Fig. 9. Direct similarity between 3D-GS and 4D-GS predictions. The high cross-model PCC shows that the two methods usually produce closely related waveforms, while 4D-GS has slightly higher predicted amplitude.

6.7 Discussion

The complete experiment provides evidence that the 4D representation is useful but not decisively superior. The strongest reproducible effect is reduced amplitude attenuation. This is valuable because the loss-function ablation showed that simply adding an explicit STD penalty to 3D-GS increased amplitude at the cost of correlation and error. In contrast, 4D-GS improves average amplitude ratios while also slightly improving PCC, MAE, and RMSE. This suggests that model representation can address part of the amplitude problem more effectively than a direct auxiliary loss.

At the same time, the limited R^2 improvement and the mixed subject-level win pattern show that representation alone does not solve the reconstruction problem. EEG channels differ in local spatial support, noise, signal amplitude, and temporal dynamics. A single fixed set of hyperparameters may therefore be suboptimal across all subjects and locations. The final 4D-GS result should be interpreted as a promising engineering extension whose mean behavior is better under the tested protocol, while additional tuning, repeated seeds, and alternative hidden-electrode sets are required before making a stronger general claim.

7 Problems Encountered and Solutions

7.1 Amplitude Attenuation

The first major modeling problem was that reconstructed signals often had substantially lower variance and RMS than the real channels. Standard MSE can favor smooth predictions when uncertainty is high, so the 3D-GS model could obtain reasonable temporal correlation while underestimating amplitude. The project tested an explicit STD loss, Huber loss, scale regularization, and Laplacian smoothing. STD loss improved the amplitude ratio but degraded the primary accuracy metrics; the other variants did not improve the trade-off. The engineering decision was therefore to

retain MSE and investigate a richer representation instead. The subsequent 4D-GS experiment improved amplitude preservation without sacrificing the average error metrics.

7.2 Validation Leakage

A missing-channel experiment is invalid if the target channel appears anywhere in the training data. To prevent this, the project moved from generating only new synthetic positions to a strict leave-electrodes-out protocol. The five target channels are removed before the training points and batches are constructed, and their data are used only after inference for evaluation. This creates a direct ground-truth test and makes the comparison between model variants auditable.

7.3 FIF Overwrite and Data Integrity

During development, modifying FIF objects in place created a risk that original channels could be overwritten or transformed unintentionally when generated electrodes were appended. The final pipeline treats the original recording as immutable. A fresh copy is reloaded before export, original data are compared numerically, and only new prediction channels are appended. The output validation file is separate from the source file. This design turns data preservation from an assumption into a tested property of the software.

7.4 Electrode-Specific Instability

Electrode 57 behaved differently from the other four validation locations, especially for 4D-GS. Rather than removing it from the reported result, the project kept it in the primary analysis and added a clearly labeled sensitivity analysis without it. This approach preserves the integrity of the predefined validation set while providing a diagnostic view of the outlier. The result also motivates future analysis of spatial neighborhood density and electrode-specific signal statistics.

7.5 Computational Scope and Project Prioritization

Training separate models over long EEG recordings and multiple subjects is computationally expensive. Extending the project simultaneously to real-time visualization, NBF benchmarking, 3D-to-4D development, loss-function tuning, and EEG-to-image reconstruction would have weakened the validation of the central reconstruction task. The scope was therefore revised: the team prioritized a complete 18-subject paired comparison and explicit acceptance tests. This resulted in a narrower but better-supported final engineering outcome.

8 Conclusions and Recommendations

8.1 Final Conclusions

The project successfully implemented an EEG-specific Gaussian reconstruction pipeline and validated it using real channels that were withheld from training. Standard MSE was selected after a

controlled S01 loss-function study. The final paired experiment on 18 subjects shows that 4D-GS has a small average advantage over 3D-GS in all six reported metrics, with the clearest improvement in amplitude preservation. Mean PCC improves by 3.24%, mean MAE and RMSE improve by less than 1%, and STD/RMS ratios improve by approximately 5.7%.

These results answer the final research question in a qualified way. Under the tested configuration, 4D-GS is preferable on average, but the improvement is not uniform across subjects or electrodes. The two models usually produce highly similar waveforms, and electrode 57 demonstrates that stronger amplitude can coexist with poorer temporal accuracy. The project therefore supports continued research on spatiotemporal Gaussian EEG models but does not claim universal 4D superiority.

8.2 Final Acceptance-Test Status

Table 7. Final acceptance-test status.

Test	Acceptance condition	Measured outcome	Status
Training separation	Five target channels absent from training	55 train / 5 held out for every paired experiment	Passed
Prediction completeness	One predicted signal per hidden channel with matching sample count	VAL_17, VAL_11, VAL_52, VAL_57, VAL_19 generated	Passed
Original-data integrity	Original channels unchanged in output construction	Source-of-truth reload and numerical comparison used	Passed
Controlled 3D/4D comparison	Same data, hidden set, preprocessing, and loss	18 paired subjects with fixed MSE	Passed
Improved mean reconstruction	4D should improve the final 3D baseline on average	4D mean winner on PCC, R^2 , MAE, RMSE, STD, RMS	Passed, modest gain
Original 10% accuracy target vs NBF	$\geq 10\%$ improvement relative to NBF	NBF was not retained as final paired baseline; 3D→4D PCC gain was 3.24%	Not demonstrated
Original 30% latency target vs NBF	$\geq 30\%$ lower training/rendering latency	No controlled final NBF runtime benchmark	Not demonstrated / descoped
Real-time visualization and EEG-to-image	Demonstration of broader preliminary goals	Removed from revised final scope	Descoped

8.3 Planned Objectives versus Final Achievement

Table 8. Planned objectives versus final achievement.

Objective	Original intent	Final outcome	Assessment
Missing-channel EEG reconstruction	Learn a spatial-temporal representation for reconstructing unmeasured locations	Five held-out physical electrodes reconstructed from the remaining 55 channels	Achieved
Virtual electrodes	Generate meaningful signals where sensors are absent	Five real channels reconstructed per subject and compared with ground truth	Achieved
3D Gaussian Splatting	Transfer explicit Gaussian representation to EEG	EEG-specific 3D Gaussian field implemented and evaluated	Achieved

4D extension	Explore temporal Gaussian representation	18-subject 3D-GS/4D-GS paired evaluation completed	Achieved
Large accuracy gain	$\geq 10\%$ over preliminary baseline	Average 4D gain over 3D was smaller on primary waveform/error metrics	Partially achieved
Compute acceleration vs NBF	Demonstrate $\geq 30\%$ latency reduction	Not measured under a controlled common hardware baseline	Not verified
Real-time 3D visualization	Interactive continuous activity visualization	Removed to prioritize reconstruction validation	Not part of final scope
EEG-to-image reconstruction	Higher-level visual-content demonstration	Removed from final scope	Not part of final scope

8.4 Advantages and Limitations

The main advantages of the final system are its explicit use of electrode geometry, direct validation against known held-out signals, modularity, and preservation of the original measurement file. The Gaussian representation is interpretable at the level of spatial centers and scales, and the controlled 3D/4D experiment isolates representation effects more cleanly than changing the model and loss function simultaneously.

The main limitations are the moderate absolute correlations, substantial remaining amplitude attenuation, only 18 paired subjects, one fixed five-electrode validation set, lack of repeated random-seed statistics, and absence of a final controlled NBF runtime/accuracy baseline. The 4D implementation also requires additional hyperparameter optimization to determine when temporal modeling provides the largest benefit. These limitations define concrete next steps rather than invalidating the observed mean improvement.

8.5 Planned versus Actual Schedule and Cost

Table 9. Planned versus actual schedule and cost summary.

Project area	Planned	Actual	Comment
Infrastructure and preprocessing	Jan. 2026	Completed early project phase	Environment, FIF processing, electrode handling, and metrics established
Model development	Mar.-Apr. 2026: NBF+GS hybrid	Direct 3D-GS followed by 4D-GS extension	Architecture changed based on implementation experience
Evaluation	Apr.-May 2026: reconstruction vs baseline	Loss ablation + 18-subject paired 3D-GS/4D-GS study	Evaluation scope expanded and became the main project contribution
Visualization / higher-level tasks	May-Jun. 2026	Removed from final scope	Time redirected to controlled reconstruction tests
Finalization	Planned by late Jul. 2026	Final analysis and report completed in Aug. 2026	Extended validation and 4D comparison increased finalization time
Direct financial cost	No dedicated budget	No direct project expenditure	Used existing academic data, open-source software, local/cluster compute resources

The project remained a software research project and did not require purchase of sensors, hardware, or proprietary software. Compute time was the primary resource. The schedule changed because the final experimental question evolved from a broad hybrid system to a deeper comparison of Gaussian

representations. The final milestone was reached later than the preliminary July target, but the additional time produced the 18-subject paired dataset and electrode-level analysis used in the final conclusions.

8.6 Recommendations for Future Work

- Repeat the 3D-GS/4D-GS experiment with multiple random seeds per subject and report confidence intervals or paired statistical tests.
- Evaluate several predefined hidden-electrode sets, including spatially easy and difficult locations, rather than relying on one set of five channels.
- Investigate electrode 57 using nearest-neighbor distance, local signal correlation, noise statistics, and channel-specific normalization.
- Tune the 4D temporal representation and regularization independently of the 3D baseline while preserving a fixed final test set.
- Compare the final Gaussian methods directly with NBF and spherical-spline interpolation under the same preprocessing and hardware conditions.
- Explore losses that preserve amplitude without directly penalizing global STD, such as frequency-domain or correlation-aware objectives.
- Only after reconstruction quality is stabilized, reintroduce continuous visualization or downstream tasks as separate validated extensions.

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Appendix A - Reproducibility and Implementation Parameters

This appendix summarizes the parameters and data-flow choices required to reproduce the final experiments. The complete source-code package should be submitted together with the final project deliverables; the report intentionally avoids duplicating full program listings in the main body.

Table 10. Reproducibility and implementation parameters.

Parameter	Final setting / behavior
Dataset	Processed Brennan EEG recordings
Original channels	60
Final paired subjects	18 subjects with both 3D-GS and 4D-GS outputs

Sampling rate	120 Hz
Window duration	3 s
Validation channels	17, 11, 52, 57, 19
Training channels	55
Input sample	[x, y, z, t] + voltage
Spatial normalization	x, y, z min-max normalized to [-1, 1]
Time normalization	Normalized temporal coordinate used by the model
Voltage normalization	Z-score normalization; metrics computed after denormalization
3D-GS centers	Unique training-electrode spatial positions
3D-GS scales	Learnable axis-wise positive scales using softplus
3D-GS temporal model	MLP: 1 → hidden → hidden → K with SiLU activations
Hidden dimension	64 in the project Gaussian field configuration
Positional encoding	None for GS; identity coordinate input
Optimizer	Adam
Final objective	MSE for both 3D-GS and 4D-GS final comparison
Output channels	VAL 17, VAL 11, VAL 52, VAL 57, VAL 19
Final metrics	PCC, R ² , MAE, RMSE, STD ratio, RMS ratio

A.1 Recommended Reproduction Sequence

- Load the filtered subject FIF and its channel positions.
- Create a training view that excludes channels 17, 11, 52, 57, and 19 while retaining their true signals for evaluation only.
- Iterate over sequential 3 s windows; normalize coordinates and voltages using the same training-window rules.
- Create either the 3D-GS or 4D-GS model and train with the fixed MSE objective.
- Predict the five hidden locations for every window and concatenate the predictions in the original temporal order.
- Denormalize voltage, reload the original FIF, append only the five VAL_* channels, and verify that original channels are unchanged.
- Compute metrics against the corresponding true channels over the complete recording.
- Aggregate per-subject and per-electrode metrics only after both 3D-GS and 4D-GS outputs are available for a paired subject.

Appendix B - Software Package Structure and Key Procedures

The project software is modularized around data preparation, Gaussian model definition, training/inference, channel management, validation, and result analysis. The complete Python source files are part of the project code deliverable. The following structure documents the role of the key modules and the implementation logic relevant to the final report.

- Run_EEG_GS_Helpers.py - GaussianEEGField definition, coordinate/voltage normalization utilities, metric calculations, and common training helpers.
- Run_Model_GS_SAFE.py and validation variants - subject-level loading, window iteration, model creation, checkpointing, prediction, safe FIF export, and integrity validation.

- 4D-GS validation runner - same hidden-channel pipeline with the spatiotemporal Gaussian representation used in the final paired experiment.
- Filtering and channel-management scripts - preprocessing, channel removal/renaming, position inspection, and file validation.
- Analysis scripts - prediction-to-ground-truth metric computation and generation of overall, subject-level, electrode-level, and direct-comparison figures.

B.1 Key 3D-GS Forward-Pass Logic

In implementation terms, the 3D-GS model computes positive spatial scales and Gaussian amplitudes, evaluates a normalized spatial mixture at the query location, evaluates a temporal network at t , and returns the weighted sum of the time-dependent Gaussian amplitudes. The core mathematical behavior is given by Eqs. (1)-(3). This design avoids Fourier positional encoding and keeps the spatial component directly interpretable in terms of electrode-centered Gaussian influence.

B.2 Safe Output Logic

Before writing a validation FIF, the export code reloads the original measurement file, verifies the expected channel/sample structure, creates the predicted channels separately, appends them with distinct VAL_* names, and checks that the measured channels have not changed. This procedure should be preserved in future versions of the project, especially if additional virtual electrodes are generated.

Appendix C - Final Cost and Milestone Record

Direct project-related financial expenditure: 0 NIS. The project used existing computing resources, public/academic EEG data, and open-source Python libraries. The primary consumed resource was GPU/cluster computation time.

Final milestone record.

Milestone	Final status	Evidence
Data and preprocessing pipeline	Completed	Filtered FIF inputs and inspection scripts
3D-GS EEG model	Completed	Validation output and loss-function experiments
Leave-electrodes-out validation	Completed	Five predefined real channels hidden from training
4D-GS extension	Completed	Paired 4D validation outputs
18-subject final comparison	Completed	Overall, subject, electrode, and direct-prediction analyses
Final report and documentation	Completed	This report and accompanying source/results package